偏微分方程第四次作业

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1. 利用 Fourier 变换求解定解问题:

$$\begin{cases} u_t - a^2 u_{xx} + b u_x + cu = f(t,x), & (t,x) \in \mathbb{R}^+ \times \mathbb{R}, \\ u(0,x) = \varphi(x), & x \in \mathbb{R}, \end{cases}$$

其中 $a \in \mathbb{R}^+, b, c \in \mathbb{R}$ 为常数.

解答: 方程两边作用 Fourier 变换可得

$$\begin{cases} \widehat{u}_t + a^2\xi^2 \widehat{u} + \mathrm{i}b\xi\widehat{u} + c\widehat{u} = \widehat{f}(t,\xi), & (t,\xi) \in \mathbb{R}^+ \times \mathbb{R}, \\ \widehat{u}(0,\xi) = \widehat{\varphi}(\xi), & \xi \in \mathbb{R}. \end{cases}$$

故

$$\widehat{u}(t,\xi) = \mathrm{e}^{-(a^2\xi^2 + \mathrm{i}b\xi + c)t}\widehat{\varphi}(\xi) + \int_0^t \mathrm{e}^{-(a^2\xi^2+\mathrm{i}b\xi+c)(t-s)}\widehat{f}(s,\xi)\,\mathrm{d}s.$$

其中

$$(\mathrm{e}^{-(a^2\xi^2+\mathrm{i}b\xi+c)t})^\vee = \mathrm{e}^{-ct}(\mathrm{e}^{-\mathrm{i}bt\xi}\mathrm{e}^{-a^2t\xi^2})^\vee = \frac{1}{\sqrt{2a^2t}}\mathrm{e}^{-ct-\frac{(x-bt)^2}{4a^2t}},$$

从而两边再作用 Fourier 逆变换可得

$$\begin{aligned} & u(t,x) \\ &= \frac{1}{\sqrt{4a^2\pi t}}\mathrm{e}^{-ct-\frac{(x-bt)^2}{4a^2t}} * \varphi(x) + \int_0^t \frac{1}{\sqrt{4a^2\pi(t-s)}}\mathrm{e}^{-c(t-s)-\frac{(x-b(t-s))^2}{4a^2(t-s)}} * f(s,x)\,\mathrm{d}s \\ &= \int_{\mathbb{R}} \frac{1}{\sqrt{4a^2\pi t}}\mathrm{e}^{-ct-\frac{(x-y-bt)^2}{4a^2t}}\varphi(y)\,\mathrm{d}y + \int_0^t\int_{\mathbb{R}} \frac{1}{\sqrt{4a^2\pi(t-s)}}\mathrm{e}^{-c(t-s)-\frac{(x-y-b(t-s))^2}{4a^2(t-s)}} f(s,y)\,\mathrm{d}y\,\mathrm{d}s. \end{aligned}$$

□

2. 利用 Fourier 变换求解定解问题:

$$\begin{cases} u_{tt} + u_{xx} = 0, & (t,x) \in \mathbb{R}^+ \times \mathbb{R}, \\ u(0,x) = \varphi(x), & x \in \mathbb{R}, \end{cases}$$

其中 $\varphi(x)$ 连续有界, 求此问题的有界解.

解答: 方程两边作用 Fourier 变换可得

$$\begin{cases} \widehat{u}_{tt} - \xi^2 \widehat{u} = 0, & (t,\xi) \in \mathbb{R}^+ \times \mathbb{R}, \\ \widehat{u}(0,\xi) = \widehat{\varphi}(\xi), & \xi \in \mathbb{R}. \end{cases}$$

故

$$\widehat{u}(t,\xi) = \begin{cases} A(\xi)\mathrm{e}^{\xi t} + B(\xi)\mathrm{e}^{-\xi t}, & \xi \neq 0, \\ A(\xi)t + B(\xi), & \xi = 0. \end{cases}$$

设 $|u(t,x)| \leqslant M$, 则 $\forall\, h \in \mathcal{S}(\mathbb{R})$, 都有

$$|\langle \widehat{u}(t), h\rangle| = |\langle u(t), \widehat{h}\rangle| \leqslant M\|\widehat{h}\|_{L^1}.$$

任取 $\psi \in C_0^\infty(\mathbb{R})$, 且 $\operatorname{supp}\psi \subseteq [a,b] \subseteq (0,+\infty)$, 并令 $h_t(\xi) = \mathrm{e}^{-\xi t}\psi(\xi)$, 则

$$\langle \widehat{u}(t), h_t\rangle = \left\langle A\mathrm{e}^{\xi t} + B\mathrm{e}^{-\xi t}, \mathrm{e}^{-\xi t}\psi\right\rangle = \langle A, \psi\rangle + \left\langle B, \mathrm{e}^{-2\xi t}\psi\right\rangle.$$

注意到 $\left\langle B, \mathrm{e}^{-2\xi t}\psi\right\rangle, \|\widehat{h}_t\|_{L^1} \xrightarrow{t\to\infty} 0$, 从而 $\langle A, \psi\rangle = 0$. 由 ψ 的任意性可得 $A \equiv 0, \forall\, \xi > 0$. 同理 $B \equiv 0, \forall\, \xi < 0$. 再代入初值在形式上可得 $\widehat{u}(t,\xi) = \widehat{\varphi}(\xi)\mathrm{e}^{-t|\xi|}$. 同时

$$(\mathrm{e}^{-t|\xi|})^\vee = \frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}} \mathrm{e}^{-t|\xi|}\mathrm{e}^{\mathrm{i}x\xi}\,\mathrm{d}\xi = \frac{1}{\sqrt{2\pi}}\left(\frac{1}{t-\mathrm{i}x} + \frac{1}{t+\mathrm{i}x}\right) = \sqrt{\frac{2}{\pi}}\frac{t}{t^2+x^2},$$

因此作用 Fourier 逆变换可得

$$u(t,x) = \frac{1}{\pi}\frac{t}{t^2+x^2} * \varphi(x) = \int_{\mathbb{R}} \frac{1}{\pi}\frac{t}{t^2+(x-y)^2}\varphi(y)\,\mathrm{d}y.$$

□

3. 求解拟线性抛物方程的初值问题

$$\begin{cases} u_t - a^2\Delta u + b\left|\nabla u\right|^2 = 0, & (t,\boldsymbol{x}) \in \mathbb{R}^+ \times \mathbb{R}^n, \\ u(0,\boldsymbol{x}) = \varphi(\boldsymbol{x}), & \boldsymbol{x} \in \mathbb{R}^n, \end{cases}$$

其中 $a \in \mathbb{R}^+, b \in \mathbb{R}$ 为常数.

解答:

(1) 若 $b \neq 0$, 令 $v := \mathrm{e}^{-\frac{b}{a^2}u}$, 则

$$v_t = -\frac{b}{a^2}\mathrm{e}^{-\frac{b}{a^2}u}u_t, \quad \Delta v = -\frac{b}{a^2}\mathrm{e}^{-\frac{b}{a^2}u}\Delta u + \frac{b^2}{a^4}\mathrm{e}^{-\frac{b}{a^2}u}\left|\nabla u\right|^2,$$

故

$$\begin{cases} v_t - a^2\Delta v = 0, & (t, \boldsymbol{x}) \in \mathbb{R}^+ \times \mathbb{R}^n, \\ v(0, \boldsymbol{x}) = \mathrm{e}^{-\frac{b}{a^2}\varphi(\boldsymbol{x})}, & \boldsymbol{x} \in \mathbb{R}^n. \end{cases}$$

从而

$$v(t, \boldsymbol{x}) = \frac{1}{(4\pi a^2 t)^{n/2}}\mathrm{e}^{-|\boldsymbol{x}|^2/4a^2t} * \mathrm{e}^{-\frac{b}{a^2}\varphi(\boldsymbol{x})} = \int_{\mathbb{R}^n} \frac{1}{(4\pi a^2 t)^{n/2}}\mathrm{e}^{-|\boldsymbol{x}-\boldsymbol{y}|^2/4a^2t}\mathrm{e}^{-\frac{b}{a^2}\varphi(\boldsymbol{y})}\,\mathrm{d}\boldsymbol{y},$$

因此

$$u(t, \boldsymbol{x}) = -\frac{a^2}{b}\ln\left(\int_{\mathbb{R}^n} \frac{1}{(4\pi a^2 t)^{n/2}}\mathrm{e}^{-|\boldsymbol{x}-\boldsymbol{y}|^2/4a^2t}\mathrm{e}^{-\frac{b}{a^2}\varphi(\boldsymbol{y})}\,\mathrm{d}\boldsymbol{y}\right).$$

(2) 若 $b = 0$, 则

$$u(t, \boldsymbol{x}) = \int_{\mathbb{R}^n} \frac{1}{(4\pi a^2 t)^{n/2}}\mathrm{e}^{-|\boldsymbol{x}-\boldsymbol{y}|^2/4a^2t}\varphi(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}.$$

□

4. 利用 Poisson 公式推导半无界问题

$$\begin{cases} u_t - u_{xx} = 0, & t > 0, x > 0, \\ u(0, x) = 0, & x > 0, \\ u(t, 0) = g(t), & t > 0, \end{cases}$$

的显式解

$$u(t, x) = \frac{x}{\sqrt{4\pi}}\int_0^t \frac{1}{(t-s)^{3/2}}\mathrm{e}^{-\frac{x^2}{4(t-s)}}g(s)\,\mathrm{d}s,$$

其中 $g \in C^1([0, +\infty))$ 且 $g(0) = 0$.

解答: 令 $v(t,x) := u(t,x) - g(t)$, 则

$$\begin{cases} v_t - v_{xx} = -g'(t), & t > 0, x > 0, \\ v(0,x) = 0, & x > 0, \\ v(t,0) = 0, & t > 0. \end{cases}$$

将 $v(t,x)$ 对 x 奇延拓到 $\mathbb{R}$ 得到 $\widetilde{v}(t,x)$, 则

$$\begin{cases} \widetilde{v}_t - \widetilde{v}_{xx} = -g'(t)\operatorname{sgn}(x), & (t,x) \in \mathbb{R}^+ \times \mathbb{R}, \\ \widetilde{v}(0,x) = 0, & x \in \mathbb{R}. \end{cases}$$

故

$$\begin{aligned} v(t,x) &= -\int_0^t \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi(t-s)}} \mathrm{e}^{-\frac{(x-y)^2}{4(t-s)}} g'(s)\operatorname{sgn}(y)\,\mathrm{d}y\,\mathrm{d}s \\ &= -\int_0^t \int_{\mathbb{R}} \frac{1}{\sqrt{\pi}} \mathrm{e}^{-z^2} g'(s)\operatorname{sgn}\left(x + \sqrt{4(t-s)}z\right)\mathrm{d}z\,\mathrm{d}s \\ &= -\frac{2}{\sqrt{\pi}} \int_0^t \int_0^{x/\sqrt{4(t-s)}} g'(s)\mathrm{e}^{-z^2}\,\mathrm{d}z\,\mathrm{d}s \\ &= -\frac{2}{\sqrt{\pi}} \int_0^t g'(s) \left(\int_0^{x/\sqrt{4(t-s)}} \mathrm{e}^{-z^2}\,\mathrm{d}z \right) \mathrm{d}s \\ &= -g(t) + \frac{1}{\sqrt{4\pi}} \int_0^t \frac{x}{(t-s)^{3/2}} \mathrm{e}^{-\frac{x^2}{4(t-s)}} g(s)\,\mathrm{d}s, \end{aligned}$$

因此

$$u(t,x) = \frac{x}{\sqrt{4\pi}} \int_0^t \frac{1}{(t-s)^{3/2}} \mathrm{e}^{-\frac{x^2}{4(t-s)}} g(s)\,\mathrm{d}s.$$

□

5. 用分离变量法求解混合问题:

$$\begin{cases} u_t - u_{xx} = u, & t > 0, 0 < x \leqslant \pi, \\ u(0,x) = x(\pi - x), & 0 \leqslant x \leqslant \pi, \\ u(t,0) = u(t,\pi) = 0, & t \geqslant 0. \end{cases}$$

解答: 设

$$u(t,x) := \sum_{n=1}^{\infty} T_n(t)\sin(nx),$$

其中 $u_n(t,x) := T_n(t)\sin(nx)$ 满足方程 $u_t - u_{xx} = u$, 故 $T_n'(t) + n^2 T_n(t) = T_n(t)$, 从而 $T_n(t) = T_n(0)\mathrm{e}^{(1-n^2)t}$, 因此

$$u(t,x) = \sum_{n=1}^{\infty} T_n(0)\mathrm{e}^{(1-n^2)t}\sin(nx).$$

再代入初值条件可得

$$u(t,x) = \sum_{n=1}^{\infty} \frac{4(1-(-1)^n)}{\pi n^3}\mathrm{e}^{(1-n^2)t}\sin(nx).$$

□

6. 设 $\Omega_T := (0,T]\times\Omega$, $a>0$ 为常数. 若 $v \in C_1^2(\Omega_T)$ 满足 $v_t - a^2\Delta v \leqslant 0, \forall\,(t,\boldsymbol{x}) \in \Omega_T$, 则称 v 在 Ω_T 上是热方程的下解.

(1) 求证: $\max\limits_{\overline{\Omega_T}} v = \max\limits_{\partial_p\Omega_T} v$;

(2) 设 $\phi : \mathbb{R}\to\mathbb{R}$ 是光滑凸函数且 u 在 Ω_T 上满足热方程. 求证: $v := \phi(u)$ 在 Ω_T 上是热方程的下解;

(3) 设 u 在 Ω_T 上满足热方程. 求证: $v := a^2\left|\nabla u\right|^2 + u_t^2$ 在 Ω_T 上是热方程的下解.

解答:

(1) 即为最大值原理.

(2) 只需注意到

$$v_t - a^2\Delta v = \phi'(u)u_t - a^2\phi''(u)\left|\nabla u\right|^2 - a^2\phi'(u)\Delta u = -a^2\phi''(u)\left|\nabla u\right|^2 \leqslant 0.$$

(3) 只需注意到

$$\begin{aligned} v_t - a^2\Delta v &= -2a^4\|\nabla^2 u\|_F^2 - 2a^2\left|\nabla u_t\right|^2 + 2a^2\nabla u\cdot(\partial_t - a^2\Delta)\nabla u + 2u_t(\partial_t - a^2\Delta)u_t \\ &= -2a^4\|\nabla^2 u\|_F^2 - 2a^2\left|\nabla u_t\right|^2 \leqslant 0. \end{aligned}$$

□